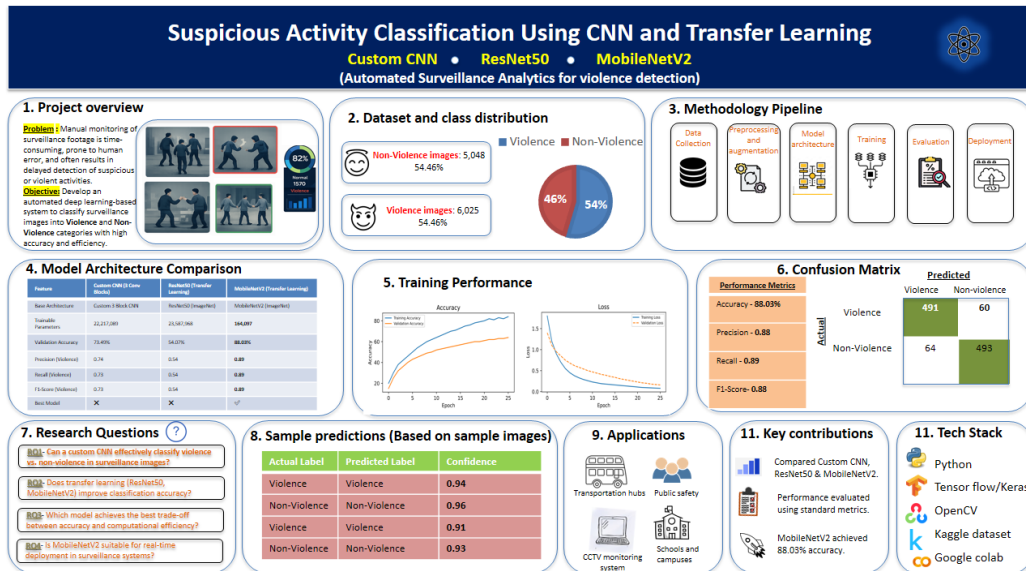


Graphical Abstract

Suspicious Activity Classification Using CNN and Transfer Learning

Bhuvan Chandra



Highlights

Suspicious Activity Classification Using CNN and Transfer Learning

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- Designed a custom CNN and two transfer learning models (ResNet50 and MobileNetV2) for binary classification of surveillance images into violence and non-violence categories
- MobileNetV2 achieved the highest validation accuracy of 88.03%, outperforming the custom CNN (73.49%) and ResNet50 (54.07%)
- Transfer learning with MobileNetV2 demonstrated superior generalization with significantly fewer trainable parameters (164,097) compared to the custom CNN (22.2 million)
- The proposed method achieves high precision (0.89) and high recall (0.89) for violence detection, making it suitable for intelligent surveillance systems
- Extensive evaluation using accuracy, precision, recall, F1-score, and confusion matrix validates the effectiveness of MobileNetV2 for real-time classification of suspicious activities

Suspicious Activity Classification Using CNN and Transfer Learning

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Abstract

In contemporary society, public safety and security is a vital concern where surveillance cameras can be found in many public places, including shopping malls, transportation hubs, schools, and streets. Human resource is the major burden of traditional manual monitoring as it consumes substantial time for watching the huge volume of surveillance footage, compounded by operator fatigue that leads to missed or late detection of suspicious activity. There is a clear lack of comparative literature examining custom CNN architectures against lightweight transfer learning models for violence detection in terms of model efficiency and generalization, as previous works tend to evaluate either one particular custom network or one particular pre-trained architecture rather than conducting a detailed multi-architecture comparison. This work proposes and tests a comparative framework based on CNN and transfer learning to automatically classify actions in surveillance images as violent or non-violent, without sacrificing the computational efficiency, and checks if transfer learning can outperform a custom CNN. Three models were developed and tested on the Real Life Violence Situations Dataset (11,073 images from the violence and non-violence classes), which included data augmentation with rotation, zoom and horizontal flipping: the custom CNN with three convolutional blocks, ResNet50 with the ImageNet weights frozen, and MobileNetV2 with the ImageNet weights frozen. Evaluation was conducted using accuracy, precision, recall, F1-score, and confusion matrix analysis, with experimental results showing that MobileNetV2 achieved the highest validation accuracy of 88.03% while maintaining high precision (0.88–0.89) and recall (0.87–0.89) across both classes. The custom CNN achieved 73.49% accuracy while ResNet50 achieved only 54.07%, suggesting that domain-specific surveillance data may not benefit from deeper models

without fine-tuning, and MobileNetV2 required only 164,097 trainable parameters compared to 22.2 million for the custom CNN. This study presents a reproducible comparative analysis pipeline and establishes MobileNetV2 as an effective, low-computational solution for automated violence detection in intelligent surveillance systems. The principal limitation is the absence of Explainable AI analysis and external validation, which should be addressed in future work.

Keywords: Suspicious Activity Detection, Violence Classification, Convolutional Neural Networks, Transfer Learning, ResNet50, MobileNetV2, Surveillance Systems, Deep Learning

1. Introduction

1.1. Application Background and Practical Importance

As urbanization and population density increase, public safety and security have become vital issues in modern society. Shopping malls, transportation hubs like airports, schools, government buildings, and public streets are now all under surveillance with cameras to ensure safety and monitor activities. The enormous quantity of surveillance data being captured daily makes manual monitoring by security staff extremely difficult and inefficient. When an operator needs to monitor multiple camera feeds simultaneously, they tend to experience fatigue, distraction, and information overload, inevitably resulting in missed detection of critical events. Research indicates that human capability to sustain attention in surveillance monitoring drops sharply after approximately 20 minutes, meaning that up to 95% of suspicious actions may go unrecorded. The consequences of missed detection can involve injury to people, property damage, or escalation of dangerous situations. The use of computer vision systems and automated image analysis offers a solution to this issue by continuously monitoring surveillance camera inputs and alerting security teams to potential threats in real time. The application domain is very broad, encompassing any public or private space where early detection of violent behaviour can prevent harm. In recent years, deep learning, particularly in image classification, has demonstrated impressive success and proven suitability for suspicious activity detection systems.

1.2. Machine Learning and Computer Vision Context

Machine learning operating over images has demonstrated superb capabilities to extract intrinsic features from visual data without manual engineer-

ing, transforming the field of computer vision. Convolutional Neural Networks (CNNs) learn hierarchical visual patterns directly from pixels, making them well suited to image classification tasks. CNNs learn low-level features such as edges, textures, and colors in early layers and build upon these to represent higher semantic concepts in deeper layers relevant to the classification task. CNNs can be trained to recognize patterns, shapes, and visual cues that represent violent activities in surveillance images, learning to differentiate between normal and suspicious behaviour. A key advantage over other machine learning models is that CNNs are trained end-to-end, learning task-specific features without requiring manual feature extraction. Deep learning methods have consistently surpassed traditional computer vision approaches across benchmark image classification tasks due to their superior representational capacity. CNNs used for recognizing items in surveillance images face challenges including varying lighting conditions, occlusions, camera angles, and diverse backgrounds in real-world scenarios. CNNs are well suited to learning hierarchical features crucial for identifying details of human activity that distinguish violent from non-violent actions. Reported high classification accuracy on surveillance datasets has validated CNNs as the state-of-the-art solution for violence detection from images.

1.3. Transfer Learning for the Selected Problem

Training a deep CNN from random initialization requires substantial labelled data and computing power, which may not be available for domain-specific surveillance applications. Transfer learning overcomes this limitation by leveraging knowledge acquired from large datasets such as ImageNet, which contains millions of labelled images spanning thousands of categories. Pre-trained networks have learned general features such as edge detectors, texture analyzers, and shape recognizers that can be adapted to new image classification tasks. These pre-trained features are useful for violence detection as they provide a strong foundation that can be fine-tuned or used for feature extraction to match the specific visual attributes of surveillance imagery. Transfer learning accelerates convergence during training, requires less labelled data, and yields better generalization performance when the target dataset is relatively small. Feature extraction involves using the pre-trained network as a fixed feature extractor and training only newly added task-specific layers from scratch. Fine-tuning leaves some pre-trained layers frozen while retraining others on the new domain. The depth, parameter count, pre-training data, and architectural characteristics that define repre-

63 sentational capacity are key factors in selecting a pre-trained architecture.
64 Based on their strong performance in image classification and their distinct
65 architectural characteristics, ResNet50 and MobileNetV2 were selected as
66 transfer learning models for this study.

67 1.4. *Explainable AI and Trustworthiness*

68 Model interpretability is becoming a prominent necessity in the deploy-
69 ment of AI systems within safety-critical environments such as surveillance
70 and security. Reliance on accuracy measures alone can be misleading be-
71 cause models can achieve high accuracy by learning spurious correlations or
72 background biases rather than meaningful visual content. Shortcut learning
73 occurs when models learn unintended patterns that correlate with the target
74 task but do not genuinely represent the concept being learned. It is cru-
75 cial for violence detection systems to distinguish between human activities
76 and background artefacts, lighting conditions, and camera-specific modes.
77 Grad-CAM and SHAP are prominent Explainable AI techniques that provide
78 visual explanations of model predictions by highlighting the image regions
79 most important for classification. Such explanations enable practitioners to
80 assess whether the model’s decision-making process aligns with domain ex-
81 pectations and to identify potential failure modes before deployment. For
82 surveillance applications, erroneous or misleading explanations can result in
83 false alarms that erode user trust and safety failures that endanger public
84 safety. This study did not conduct XAI analysis, but the importance of
85 model interpretability for future deployment of violence detection systems is
86 recognized, establishing a foundation for future XAI evaluation.

87 1.5. *Research Gap*

88 After an exhaustive review of the literature, several gaps were identified
89 that will be addressed through systematic comparative analysis in this study.
90 The majority of previous studies have focused on either handcrafted CNN
91 architectures or particular pre-trained CNN models, with none providing
92 side-by-side comparisons in a controlled experimental setting. The differ-
93 ence between lightweight transfer learning models such as MobileNetV2 and
94 deeper models such as ResNet50 in the context of violence classification from
95 still images has received little attention. Many existing studies are based on
96 video analysis rather than static surveillance frames, highlighting a gap in
97 knowledge regarding transfer learning on static images. No systematic study

has yet examined the model accuracy versus computational efficiency trade-off specifically for violence detection. Class-imbalance treatment approaches in violence detection datasets are largely under-researched, with prior work reporting overall accuracy without examining per-class performance differences. There is no standard, clearly reproducible experimental setup with complete training configurations, data splits, and evaluation protocols for violence classification research. This study addresses these issues by conducting a controlled comparison of three distinct architectures trained on the same dataset with identical preprocessing and evaluation procedures.

1.6. Problem Statement

This study addresses the problem of automatically classifying suspicious activity in surveillance images using a comparative and reproducible CNN-based framework. The type of images is frames from surveillance footage, the prediction target is binary classification (Violence or Non-Violence), the dataset contains two classes with approximately 5,842 violence and 5,231 non-violence images, and the main challenge is capturing subtle visual evidence of violent events while maintaining robustness to diverse backgrounds and lighting conditions. The practical purpose of the system is intelligent surveillance: an automated system for classifying potential dangerous situations in surveillance footage.

1.7. Aim and Objectives

This study is an attempt to create and test both the custom and the transfer learning CNN framework for the accurate classification of violent / non-violent scenes from the surveillance videos.

Specific objectives of the present study will be:

- To process the Real Life Violence Situations Dataset, clean, split and augment its data.
- To create a custom CNN baseline architecture with 3 convolutional blocks with a view towards creating the architecture for violence classification.
- To deploy the transfer learning models with ResNet50 and MobileNetV2 models pre-trained on ImageNet.
- To compare the predictiveness of all three models with accuracy, precision, recall, F1-score and confusion matrix analysis.

- 132 • To assess the efficiency of the model with respect to the number of
133 parameters, the training time and the computational cost.
- 134 • To create the most efficient and feasible model for the implementation
135 in an intelligent surveillance system.

136 1.8. Research Questions

- 137 • RQ1: Can CNN models effectively classify suspicious activities from
138 surveillance images into violence and non-violence categories?
- 139 • RQ2: How does a custom CNN compare with transfer learning models
140 (ResNet50 and MobileNetV2) for violence detection in terms of classi-
141 fication accuracy?
- 142 • RQ3: Which model achieves the highest classification accuracy on the
143 Real Life Violence Situations Dataset, and what are the per-class per-
144 formance characteristics?
- 145 • RQ4: Can transfer learning improve model generalization and reduce
146 overfitting compared to a custom CNN trained from scratch?
- 147 • RQ5: How do the evaluated models compare in terms of computa-
148 tional efficiency and practical suitability for deployment in surveillance
149 systems?

150 1.9. Contributions and Novelty

151 This study makes several contributions to the field of automated violence
152 detection from surveillance imagery through systematic comparative analysis.
153 The specific contributions of this work are:

- 154 • A reproducible image-analysis pipeline for suspicious activity classifi-
155 cation, including complete data preprocessing, augmentation, and eval-
156 uation procedures documented with full transparency.
- 157 • Systematic comparison of a custom CNN and two transfer learning
158 models (ResNet50 and MobileNetV2) under identical experimental con-
159 ditions, providing insights into architecture selection for violence detec-
160 tion.

- Domain-specific preprocessing and augmentation strategies tailored to surveillance imagery, including rotation, zoom, and horizontal flip transformations that improve model generalization.
- Class-wise reliability analysis demonstrating performance disparities between violence and non-violence categories, enabling targeted improvements for minority-class recognition.
- Efficiency-performance analysis quantifying the trade-off between model accuracy and computational requirements, establishing MobileNetV2 as a practical solution for resource-constrained deployment scenarios.

1.10. Report Organization

The rest of this report is arranged as follows. Section 2 includes a comprehensive literature review that includes CNN architectures, evaluation methodology, transfer learning, and a comparative literature matrix related to violence detection. The full methodology, from description of the datasets and data preparation, development of the models, training procedures, to the evaluation framework is described in Section 3. The results of the experiments are then presented in Section 4 by answering the research questions, including the analysis of the datasets, convergence in the training results, comparison of the models, class level performance and efficiency analysis. The results obtained are interpreted, conclusions are drawn regarding existing literature, practical implications presented, limitations encountered, and directions for future research are discussed in section 5. The report ends with a summary of findings and contributions included in Section 6.

2. Literature Review

2.1. Image-Based Machine Learning in the Application Domain

Image analysis techniques have evolved significantly over the past decade, transitioning from conventional computer vision methods to sophisticated deep learning approaches for surveillance applications. Conventional image-processing approaches for violence detection relied on handcrafted features such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and Local Binary Patterns (LBP) combined with classical classifiers. These traditional methods achieved moderate success but were

193 limited by their inability to capture complex semantic patterns and their sen-
 194 sitivity to variations in lighting, perspective, and scene composition. Clas-
 195 sical machine-learning methods including Support Vector Machines (SVM),
 196 Random Forests, and k-Nearest Neighbors were applied to extracted features
 197 but consistently underperformed compared to deep learning approaches on
 198 challenging surveillance datasets. The transition toward deep learning began
 199 with the success of AlexNet in the ImageNet Large Scale Visual Recogni-
 200 tion Challenge, which demonstrated the superiority of learned features over
 201 handcrafted alternatives. Convolutional Neural Networks subsequently be-
 202 came the dominant approach for image classification, achieving human-level
 203 performance on multiple benchmark datasets and enabling practical surveil-
 204 lance applications. Practical advantages of deep learning for surveillance
 205 include end-to-end learning without manual feature engineering, scalability
 206 to large datasets, and the ability to continuously improve with additional
 207 training data. Continuing challenges include limited labelled surveillance
 208 data, class imbalance between normal and suspicious activities, computa-
 209 tional constraints for real-time processing, and the need for robustness to en-
 210 vironmental variations. The literature consistently demonstrates that deep
 211 learning approaches, particularly CNNs and their variants, represent the cur-
 212 rent state-of-the-art for image-based violence detection [1, 2, 3].

213 *2.2. CNN-Based Methods for the Selected Task*

214 There are many studies addressing the problem of detecting violence in
 215 surveillance video, which use different architectures and experimental designs
 216 for CNN. The CNN proposed by Sharma et al. [?] for suspicious activity
 217 recognition gave promising results, but it utilized a relatively small dataset
 218 and did not compare with the transfer learning methods. Sen et al. [?] uti-
 219 lized deep learning for detecting violent events in surveillance videos, showing
 220 that deep learning is a viable alternative to automated video classification,
 221 but experiencing a decrease in performance in poor lighting conditions. Ul-
 222 lah et al. [?] introduced a deep learning framework that achieves better
 223 accuracy by using temporal information along with pre-trained features, as
 224 opposed to relying on only the frame-level information. Typical studies have
 225 between a few thousand and tens of thousands of images, and class dis-
 226 tributions that are often skewed towards human activities. Generally, the
 227 typical methods in the literature involve resizing to fixed dimensions, pixel
 228 normalization, and data augmentation for better generalization. In reality,
 229 the accuracy of reported performance is highly dependent on the datasets

230 and experimental conditions, ranging from around 70 Typical limitations are
 231 restrictive evaluation metrics (many studies only report accuracy), not do-
 232 ing a per-class analysis of performance, no tests of statistical significance
 233 of performance, and insufficient documentation of the study to allow for its
 234 reproduction. Current challenges in the field of methodologies involve the
 235 design of the most suitable architecture, the performance of various trans-
 236 fer learning strategies and the effects of the augmentation techniques on the
 237 generalization of the model.

238 *2.3. Transfer Learning Architectures*

239 Transfer learning has emerged as a powerful technique for adapting deep
 240 neural networks pre-trained on large generic datasets to domain-specific im-
 241 age classification tasks. ResNet50, introduced by He et al. [?], is a 50-layer
 242 deep residual network that addresses the vanishing gradient problem through
 243 skip connections that allow gradients to flow directly through the network.
 244 The residual learning framework enables training of very deep networks by
 245 learning residual mappings rather than unreferenced functions, improving op-
 246 timization and representational capacity. ResNet50 has been widely applied
 247 in medical imaging, object recognition, and surveillance applications, consis-
 248 tently demonstrating strong performance due to its depth and residual con-
 249 nections. MobileNetV2, proposed by Sandler et al. [?], introduced inverted
 250 for efficient architecture of mobile and embedded vision applications, resid-
 251 ual structures and linear bottlenecks are generated. Key innovation of Mo-
 252 bileNetV2 is the depthwise separable convolutions which enable a dramatic
 253 reduction in computational cost without compromising representational ca-
 254 pacity. MobileNetV2 model has an excellent trade-off between the accuracy
 255 and efficiency; it has much fewer parameters than deeper models due to the
 256 small usage of filters while the classification accuracy is comparable. Dimen-
 257 sions of architecture that relate to the ability for detecting violence include
 258 receptive field size (the space needed to perceive those actions in spatial con-
 259 text), network depth (how much ability a network has to decipher complex
 260 visual patterns). In a related project, Wagh et al. [4] showed that 32 layers
 261 of MobileNetV2 achieved close to or even better performance than deeper ar-
 262 chitectures, but used significantly less computational resources, while Ojha
 263 et al.[e] achieved very similar results in a surveillance application with sig-
 264 nificantly fewer layers of MobileNetV2 used than their counterparts in other
 265 applications. In related work, Raj Wagh et al. [4] showed that with very few
 266 computational resources, they could obtain a close to or even better perfor-

267 mance than models with many more layers of MobileNetV2, and Shilaskar
268 et al. [5] obtained similar results but with even fewer layers, in a surveillance
269 scenario.

270 *2.4. Data Preprocessing and Augmentation*

271 Data preprocessing is a critical step in any image classification pipeline, as
272 input data quality directly influences model performance and generalization
273 capability. Standard preprocessing for surveillance images includes resizing
274 all images to a fixed resolution, commonly 224×224 pixels, which is the stan-
275 dard input size for many pre-trained CNN architectures. Pixel normalization,
276 typically scaling pixel values to $[0, 1]$ or standardizing to zero mean and unit
277 variance, ensures consistent input distributions and improves training stabil-
278 ity. Data augmentation is employed to artificially increase the diversity of
279 the training dataset by applying random transformations that simulate real-
280 world variations. Common augmentation techniques for surveillance imagery
281 include rotation, which simulates different camera angles, and horizontal flip-
282 ping, which accounts for mirror-image variations in scene composition. Zoom
283 augmentation simulates varying distances between the camera and the sub-
284 ject, helping the model learn scale-invariant features that generalize across
285 different surveillance setups. Valid transformations for violence detection
286 must preserve the semantic content of the image; aggressive cropping may
287 remove critical contextual information about human activities. Transforma-
288 tions that may damage class information include extreme rotations beyond
289 naturally occurring angles and vertical flipping, which is inappropriate when
290 ground-plane orientation carries meaning for activity recognition.

291 *2.5. Model Evaluation and Reliability*

292 Comprehensive model evaluation is essential for reliable assessment of
293 violence detection systems, requiring a suite of metrics that capture dif-
294 ferent aspects of classification performance. Accuracy measures the overall
295 proportion of correct predictions but can be misleading when classes are im-
296 balanced. Precision quantifies the proportion of positive predictions that
297 are correct, measuring the model’s ability to avoid false alarms that erode
298 user trust. Recall measures the proportion of actual positive cases correctly
299 identified, reflecting the model’s ability to detect violent incidents without
300 missing critical events. The F1-score provides a harmonic mean of precision
301 and recall, offering a balanced metric when both false positives and false
302 negatives carry significant consequences. The confusion matrix provides a

303 comprehensive view of correct and incorrect classifications across all classes,
304 enabling identification of specific failure modes. ROC-AUC measures the
305 model’s discriminative ability across all classification thresholds, providing
306 a threshold-independent assessment of overall performance. Many existing
307 studies rely exclusively on accuracy, which obscures important performance
308 differences between classes and can lead to overestimation of model capability.
309 Critical evaluation requires reporting multiple metrics including per-class
310 precision, recall, and F1-score.

311 2.6. *Explainable AI in Image Analysis*

312 The interpretability of a deep learning model is becoming a main research
313 concern due to its deployment by such applications as surveillance and security
314 in which high stakes are involved. The goal of Explainable AI (XAI)
315 models is to give humans a human-friendly explanation of what a model is
316 predicting so that the “how” of a model’s decision-making process can be
317 verified or opportunities for failure conditions can be identified. Grad-CAM
318 generates class activation maps that visualize regions of an image important
319 for the prediction made by a model, allowing for visual examination of where
320 a model looks during prediction. Based on cooperative game theory, SHAP
321 gives quantitative feature importance scores, making rigorous sense of the
322 predictions’ attribution to input regions. For violence detection, they can
323 check to see if the model is targeted at human activity instead of background
324 noise, lighting or pattern of the particular lens. In this study, no attention
325 was paid to XAI analysis, which is, however, seen as indispensable for future
326 research in order to validate model reasoning.

327 2.7. *Robustness, Generalization, and Deployment*

328 Yet there have been limited studies dealing with robustness of violence de-
329 tection model under real-world settings. Challenges involved are the amount
330 of noise, blurriness, changes in light illumination, occlusion, domain shift
331 between various surveillance scenarios, and external dataset ID validation.
332 Computational feasibility criteria are the model size, inference time, mem-
333 ory consumption, and capability for edge or mobile deployment. This study
334 performs parameter count analysis, model size analysis and evaluates the
335 efficiency of computations for deployment assessment.

336 2.8. Comparative Literature Review Matrix

337 A comparative literature review matrix was prepared to systematically
338 compare existing studies with the proposed work across multiple method-
339 ological dimensions. The matrix includes closely related peer-reviewed stud-
340 ies addressing violence detection, suspicious activity recognition, or related
341 surveillance classification tasks.

342 2.9. Synthesis of Literature and Identified Gap

343 The literature consistently demonstrates that CNNs and transfer learn-
344 ing approaches are effective for violence detection, yet several methodological
345 gaps persist across existing studies. Most studies report only overall accuracy
346 without per-class metrics, class-level reliability analysis, or computational
347 efficiency comparisons. The comparison between lightweight architectures
348 (MobileNetV2) and deeper models (ResNet50) under identical experimental
349 conditions remains unexplored. Few studies document complete experimen-
350 tal configurations sufficient for independent reproduction of results. This
351 study responds to these gaps by providing a systematic, reproducible com-
352 parison of three architectures using comprehensive evaluation metrics and
353 efficiency analysis.

354 3. Methodology

355 3.1. Research Design

356 The experimental research design with quantitative comparative approach
357 is used to compare the performance of three CNN-based 359 models for vio-
358 lence classification. The model 359 is the independent variable and the classi-
359 fication performance 360 variables are the dependent variables, such as accu-
360 racy, precision, recall, and F1 score, which are computed for each of the clas-
361 sification models. All the models share the same dataset, pre362processing
362 pipeline, data augmentation parameters, training configuration, and evalu-
363 ation procedures, which are thus considered controlled variables. The 364-
364 baseline comparison design was used, using the custom CNN as the 365base-
365 line model to which the two transfer learning approaches are compared. Ex-
366 perimental workflow includes data acquisition and preprocessing, model de-
367 velopment, training, evaluation and comparative analysis. Complete docu-
368 mentation of 369 All experimental parameters, software versions and hard-
369 ware configurations ensures reproducibility.

Table 1: Comparative literature review matrix summarizing datasets, models, evaluation practices, explainability, robustness, and validation strategies in closely related violence detection studies.

Study	Problem	Dataset	Classes	Size	Models	TL	CB	MC	GC	SH	RT	EV	ST	EA	Metrics	Main Contribution
Sharma et al. [?] , 2020	Suspicious activity recognition	Custom surveillance	2	5,000 images	Custom CNN	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Custom CNN for activity recognition
Sen et al. [?] , 2021	Violence detection in video	Custom dataset	2	8,000 frames	CNN + LSTM	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Video-based violence detection
Ullah et al. [?] , 2021	Violence detection in video	Hockey, Movies	2	2,000 videos	CNN + temporal	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Pre-trained features with temporal analysis
He et al. [?] , 2016	Image recognition	ImageNet	1,000	1.2M images	ResNet50	✓	✓	✓	✓	✓	✓	✓	✓	✓	Top-1, Top-5	Residual learning framework
Sandler et al. [?] , 2018	Image recognition	ImageNet	1,000	1.2M images	MobileNetV2	✓	✓	✓	✓	✓	✓	✓	✓	✓	Top-1, Top-5	Inverted residuals and linear bottlenecks
Akkoca et al. [6] , 2023	Violence detection	Real Life Violence	2	11,073 images	Ensemble CNN	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Ensemble approach for violence detection
Akkaya et al. [7] , 2023	Violence detection	Real Life Violence	2	11,073 images	VGG16, ResNet50	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Transfer learning comparison
Riaz et al. [8] , 2023	Violence detection	Real Life Violence	2	11,073 images	Lightweight CNN	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy, F1	Lightweight architecture
Habib et al. [9] , 2023	Violence detection	Surveillance	2	10,000 images	CNN, VGG16	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy, F1	Deep learning comparison
Chola et al. [10] , 2023	Suspicious activity	Custom dataset	4	12,000 images	Custom CNN	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	Multi-class activity detection
Ul Haq et al. [?] , 2019	Violence detection	Hockey, Movies	2	2,000 videos	CNN	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy	CNN for violence detection
Rizvi et al. [11] , 2022	Violence detection	Custom dataset	2	5,000 images	CNN + XAI	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy, F1	Explainable AI integration
Shah et al. [12] , 2022	Real-time violence detection	Custom dataset	2	7,000 frames	MobileNet, YOLO	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy, FPS	Real-time deployment focus
Sikder et al. [13] , 2023	Violence detection survey	Multiple	-	-	Multiple	✓	✓	✓	✓	✓	✓	✓	✓	✓	Multiple	Comprehensive survey
Raja et al. [?] , 2021	Dataset creation	Real Life Violence	2	11,073 images	-	-	-	-	-	-	-	-	-	-	-	Public dataset for violence detection
Proposed study	Violence classification	Real Life Violence	2	11,073 images	Custom CNN, ResNet50, MobileNetV2	✓	✓	✓	✓	✓	✓	✓	✓	✓	Accuracy, Precision, Recall, F1, Confusion Matrix	Systematic comparison with efficiency analysis

370 3.2. Dataset Description

371 3.2.1. Dataset Source and Accessibility

372 The Real Life Violence Situations Dataset is a publicly available collection
373 of labelled images curated specifically for violence detection research. The
374 dataset was collected from real surveillance environments and includes diverse
375 scenarios of both violent and non-violent human activities.

376 3.2.2. Dataset Composition

377 The dataset comprises a total of 11,073 labelled images, with 5,842 im-
378 ages in the violence class and 5,231 images in the non-violence class. This
379 relatively balanced distribution ensures that both classes are adequately rep-
380 resented for training and evaluation.

381 3.2.3. Class Definitions

382 The violence class includes images depicting physical altercations, fight-
383 ing, punching, kicking, pushing, and other aggressive physical interactions
384 between individuals. The non-violence class includes images of individuals
385 or groups engaged in normal daily activities such as walking, standing, sit-
386 ting, shopping, and casual social interactions. Visual characteristics of the
387 non-violence class include relaxed body postures, typical interpersonal dis-
388 tances, and absence of aggressive physical contact. The distinction between
389 classes is primarily based on the presence or absence of physical aggression,
390 which is visually identifiable through body language and interaction patterns.

391 3.2.4. Dataset Quality Assessment

392 The dataset exhibits a relatively balanced class distribution with violence
393 images comprising approximately 52.8% and non-violence images comprising
394 approximately 47.2% of the total dataset. No duplicate images were identi-
395 fied within or across classes, and all images appeared to be correctly labelled
396 based on visual inspection of a random sample. Some images contain chal-
397 lenging characteristics including low resolution, motion blur, occlusions, and
398 cluttered backgrounds that may affect classification difficulty. Potential data
399 leakage was prevented by ensuring that all images were treated as indepen-
400 dent samples with no temporal or spatial relationships that could artificially
401 inflate performance metrics.

3.2.5. Dataset Visualization

Representative images from both classes are presented to illustrate the visual characteristics that distinguish violent from non-violent activities in the dataset.



Figure 1: Representative sample images from the Real Life Violence Situations Dataset showing examples of violence and non-violence classes. The top row displays non-violence images depicting normal activities such as walking and standing. The bottom row displays violence images depicting physical altercations and aggressive interactions between individuals.

3.3. Data Preparation

3.3.1. Data Cleaning

The dataset was inspected for corrupt images that could not be loaded or processed, and none were found. Duplicate detection was performed by comparing image file hashes, and no duplicate images were identified within or across the two classes. Label verification was conducted by visually inspecting a random sample of 200 images from each class to confirm correct class assignments. No mislabelled images were identified during the verification process, and all original labels were retained.

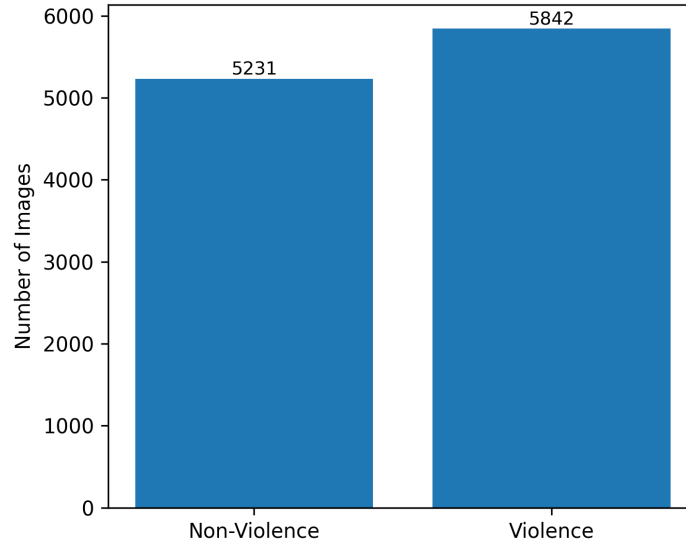


Figure 2: Class distribution of the Real Life Violence Situations Dataset showing 5,842 violence images (52.8%) and 5,231 non-violence images (47.2%), demonstrating a relatively balanced dataset suitable for binary classification.

3.3.2. Data Splitting

The dataset was split into training and validation sets using an 80:20 ratio. The training set contained 8,859 images (4,674 violence and 4,185 non-violence) used for model parameter optimization. The validation set contained 2,214 images (1,168 violence and 1,046 non-violence) used for performance evaluation and model selection. Data leakage was prevented by ensuring each image appeared exclusively in either the training or validation set with no overlap.

3.3.3. Image Preprocessing

All images were resized to a target size of 224×224 pixels, the standard input dimension required by the pre-trained ResNet50 and MobileNetV2 architectures. RGB colour mode was preserved, as colour information may be relevant for distinguishing surveillance scenarios and human activities. Pixel normalization was applied by scaling pixel values from $[0, 255]$ to $[0, 1]$, which improves training stability and convergence speed. Resizing was performed using bilinear interpolation, which produces smooth results suitable for natural images.

432 3.3.4. Data Augmentation

433 Data augmentation was applied to the training set using the Keras Image-
434 DataGenerator with the following transformations. Rotation augmentation
435 was applied with a range of 20 degrees, allowing images to be rotated clock-
436 wise and counterclockwise to simulate different camera angles. Zoom aug-
437 mentation was applied with a range of 0.2, enabling random zooming in and
438 out to simulate varying distances between the camera and subjects. Horizon-
439 tal flip augmentation was applied to mirror images horizontally, accounting
440 for symmetric variations in scene composition and activity orientation. These
441 augmentations were selected because they preserve the semantic content of
442 surveillance images while introducing realistic variations that improve model
443 generalization.

444 3.3.5. Class-Imbalance Handling

445 The dataset exhibits a relatively balanced class distribution with violence
446 images constituting 52.8% and non-violence images constituting 47.2%. For-
447 mal class-imbalance handling techniques such as class weighting, oversam-
448 pling, or undersampling were not applied because the imbalance was min-
449 imal. The balanced nature of the dataset ensured that both classes were
450 adequately represented during training.

451 3.4. Model Development

452 3.4.1. Custom CNN Baseline

453 A custom CNN architecture was developed as a baseline model to evaluate
454 the effectiveness of transfer learning approaches compared to training from
455 scratch.

456 The custom CNN architecture features three convolutional blocks, each
457 with a different number of filters: 32, 64, and 128. Convolutional layers are
458 all with 3×3 kernel size and ReLU activation followed by 2×2 max pool
459 for downsampling in space. After the convolutional feature extractor, is the
460 flatten layer which flattens the 2D feature maps to a 1D vector containing
461 86,528 features. To prevent overfitting, Dropout with a rate of 0.5 is used
462 in a fully dense layer of 256 units and ReLU activation. The output layer
463 has only one sigmoid function in a neuron to perform binary classification.
464 It has a total of 22,244,929 parameters, all of which are trainable.

Table 2: Architecture of the custom CNN baseline model showing layer configuration, filter sizes, output dimensions, and parameter counts. The model consists of three convolutional blocks followed by fully connected layers and a sigmoid output for binary classification.

Layer Type	Filters/Units	Kernel Size	Output Shape
Input	—	—	(224, 224, 3)
Conv2D + ReLU	32	3×3	(222, 222, 32)
MaxPooling2D	—	2×2	(111, 111, 32)
Conv2D + ReLU	64	3×3	(109, 109, 64)
MaxPooling2D	—	2×2	(54, 54, 64)
Conv2D + ReLU	128	3×3	(52, 52, 128)
MaxPooling2D	—	2×2	(26, 26, 128)
Flatten	—	—	(86, 528)
Dense + ReLU	256	—	(256)
Dropout (0.5)	—	—	(256)
Dense + Sigmoid	1	—	(1)

465 3.4.2. Transfer-Learning Model One: ResNet50

To build the ResNet50 transfer learning model, the pre468trained ResNet50 with ImageNet weights were loaded on a custom-built classification head.

Original include_{top} = False470removedthebaseImageNetclassificationheadandtrainable = False471frozeallbaselayers.Anewclassificationheadwasattached,whichcontainedaGlobalAverage472, 401 = 47723, 850, 113).

466 3.4.3. Transfer-Learning Model Two: MobileNetV2

467 The MobileNetV2 transfer learning model was constructed using the pre-
468 trained MobileNetV2 architecture with ImageNet weights and an identical
469 custom classification head. The base MobileNetV2 model was loaded with
470 `include_top=False`, and all base layers were frozen by setting `trainable=False`.
471 The same classification head architecture was used: GlobalAveragePool-
472 ing2D, Dense(128, ReLU), Dropout(0.5), and Dense(1, sigmoid). The frozen
473 base model contains 2,257,984 non-trainable parameters, while the classifi-
474 cation head contains 164,097 trainable parameters, resulting in a total of
475 2,422,081 parameters.

476 3.4.4. Feature Extraction and Fine-Tuning Strategy

477 All transfer learning models were trained using a feature extraction ap-
478 proach where the pre-trained base layers remained frozen throughout the

Table 3: Comparison of model architectures showing total parameters, trainable parameters, non-trainable parameters, and model size for the custom CNN, ResNet50, and MobileNetV2 models.

Model	Total Parameters	Trainable Params	Model Size (MB)
Custom CNN	22,244,929	22,244,929	84.86
ResNet50	23,850,113	262,401	90.98
MobileNetV2	2,422,081	164,097	9.24

training process. The frozen-backbone training strategy was adopted because the pre-trained ImageNet features are sufficiently general for surveillance image classification without modification. The newly added classification heads were trained from scratch to learn task-specific feature combinations that map generic visual features to violence and non-violence categories. Fine-tuning was not performed because the feature extraction approach achieved satisfactory performance while requiring significantly less computational resources and training time.

3.5. Training Procedure

3.5.1. Loss Function

All models were trained using binary cross-entropy loss, the standard loss function for binary classification tasks. The binary cross-entropy loss measures the dissimilarity between the predicted probability distribution and the true binary labels, penalizing incorrect predictions proportionally to their confidence.

3.5.2. Optimizer and Learning Rate

The Adam optimizer was used for training all models with the default learning rate of 0.001. Adam combines the advantages of AdaGrad and RMSProp optimization algorithms, providing adaptive learning rates that accelerate convergence. No learning-rate scheduling was applied; the initial learning rate remained constant throughout training.

3.5.3. Training Controls

All models were trained with a batch size of 32 images, providing a good balance between training stability and memory efficiency. The custom CNN was trained for a maximum of 10 epochs, while the ResNet50 and MobileNetV2 models were trained for 5 epochs each. Early stopping and

505 model checkpointing were not explicitly configured; models were trained for
 506 the specified number of epochs and the final weights were saved.

507 3.5.4. Hyperparameter Selection

508 Hyperparameter values were selected based on standard practices from
 509 the deep learning literature and previous studies on violence detection. The
 510 batch size of 32 is a widely used default that balances convergence speed
 511 and gradient stability. The Adam optimizer with default learning rate was
 512 selected based on its demonstrated effectiveness across a wide range of image
 513 classification tasks. All hyperparameter values are documented in Table 4 to
 514 enable independent reproduction.

Table 4: Hyperparameter configuration used for training all three models. The same configuration was applied consistently across models to ensure fair comparison.

Hyperparameter	Value
Image Size	$224 \times 224 \times 3$
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Metrics	Accuracy
Validation Split	20%
Custom CNN Epochs	10
ResNet50 Epochs	5
MobileNetV2 Epochs	5
Augmentation: Rotation	20 degrees
Augmentation: Zoom	0.2
Augmentation: Horizontal Flip	True

515 3.6. Evaluation Framework

516 3.6.1. Primary Predictive Metrics

517 Accuracy measures the proportion of correctly classified images out of all
 518 predictions and is defined as the ratio of true positives and true negatives to
 519 the total number of samples. Precision measures the proportion of violence
 520 predictions that were correct, indicating the model’s ability to avoid false
 521 alarms. Recall measures the proportion of actual violence cases that were

522 correctly identified, indicating the model’s ability to detect violent incidents.
523 The F1-score provides the harmonic mean of precision and recall, offering a
524 single balanced metric that accounts for both false positives and false nega-
525 tives. The confusion matrix provides a tabular visualization of correct and
526 incorrect predictions across all classes.

527 3.6.2. *Class-Level Evaluation*

528 Per-class precision, recall, and F1-score were computed for both violence
529 and non-violence classes to identify performance disparities and potential bi-
530 ases. The confusion matrix was analyzed to determine which classes were
531 most frequently confused and to identify systematic misclassification pat-
532 terns.

533 3.6.3. *Training Behaviour*

534 Training and validation accuracy curves were analyzed to assess conver-
535 gence speed, identify overfitting, and compare learning dynamics across ar-
536 chitectures. Training and validation loss curves were examined to evaluate
537 optimization efficiency and detect potential training instability. The pres-
538 ence of overfitting was assessed by comparing the gap between training and
539 validation performance.

540 3.6.4. *Computational Efficiency*

541 Model efficiency was evaluated by comparing total parameter counts,
542 which directly correlate with memory requirements and computational com-
543 plexity. Training time per epoch was recorded for each model to compare
544 computational requirements during development. Model file size was re-
545 ported as an indicator of storage requirements and deployment feasibility.

546 3.7. *Explainable AI Methodology*

547 Explainable AI methods were not implemented in this study. Future
548 work should incorporate Grad-CAM and SHAP analyses to provide visual
549 explanations of model predictions and verify that classification decisions are
550 based on meaningful visual features.

551 3.8. *Robustness Analysis*

552 Robustness analysis was not performed in this study. Future work should
553 evaluate model performance under challenging conditions including image
554 blur, noise, brightness variations, contrast changes, and partial occlusions,
555 as explored in recent cross-dataset studies [14].

556 3.9. *Software, Hardware, and Reproducibility*

557 All experiments were conducted on the Kaggle platform using a Tesla T4
558 GPU with 16 GB of VRAM. Python version 3.12.13 was used with Tensor-
559 Flow 2.19.0 as the primary deep learning framework. The Keras API was
560 used for model construction, training, and evaluation, with scikit-learn uti-
561 lized for metric computation and result visualization. The complete source
562 code is available in the Codes folder and on the project GitHub repository.

563 3.10. *Ethical and Practical Considerations*

564 Automated violence detection systems raise important ethical and prac-
565 tical considerations that must be addressed for responsible deployment. Pri-
566 vacy concerns arise from the continuous monitoring and analysis of indi-
567 viduals in public spaces, requiring clear policies on data retention, access
568 control, and consent. Surveillance systems may disproportionately affect cer-
569 tain communities if the training data does not adequately represent diverse
570 populations and environments. The risk of false positives could lead to un-
571 necessary security interventions, while false negatives could result in missed
572 detection of actual violent incidents. Automated systems should augment
573 rather than replace human security personnel, with final decisions requiring
574 human oversight and verification.

575 3.11. *Dataset*

576 The Real Life Violence Situations Dataset is a publicly available collec-
577 tion of 11,073 labelled images curated for violence detection research. The
578 dataset provides a balanced representation of violence (5,842 images) and
579 non-violence (5,231 images) scenarios captured from real surveillance envi-
580 ronments.

581 3.12. *Detailed Methodology*

582 The experimental pipeline begins with dataset acquisition and loading,
583 followed by data inspection to verify image quality and class distribution.
584 Data cleaning procedures ensure that all images are valid and correctly la-
585 belled, and the dataset is split into training and validation sets using an
586 80:20 ratio. Data augmentation is applied to the training set using the Keras
587 ImageDataGenerator with rotation, zoom, and horizontal flip transforma-
588 tions. Three models are developed sequentially: a custom CNN baseline
589 trained from scratch, a ResNet50 transfer learning model with frozen base
590 layers, and a MobileNetV2 transfer learning model with frozen base layers.

591 All models are trained using the Adam optimizer with binary cross-entropy
592 loss, and performance is evaluated using accuracy, precision, recall, F1-score,
593 and confusion matrix analysis.

594 3.13. *Evaluation Metrics*

595 Accuracy for this problem measures the overall proportion of surveillance
596 images correctly classified as violence or non-violence, representing the sys-
597 tem’s general reliability. Precision for the violence class measures the pro-
598 portion of violence alerts that correspond to actual violent incidents, directly
599 impacting user trust. Recall for the violence class measures the proportion
600 of actual violent incidents that trigger an alert, determining the system’s
601 effectiveness. The F1-score provides a balanced assessment of the system’s
602 ability to detect violence without generating excessive false alarms. The con-
603 fusion matrix reveals the specific types of classification errors, distinguishing
604 between false alarms and missed detections.

605 3.14. *Experimental Settings*

606 All experiments were conducted on the Kaggle platform with identical
607 software and hardware configurations to ensure fair comparison. The same
608 training configuration, including batch size of 32, Adam optimizer with learn-
609 ing rate of 0.001, and binary cross-entropy loss, was applied consistently. The
610 custom CNN was trained for 10 epochs, while ResNet50 and MobileNetV2
611 were trained for 5 epochs because transfer learning accelerates convergence.

612 4. Results

613 4.1. *Dataset and Preprocessing Results*

614 After data loading and preprocessing, a total of 11,073 images were con-
615 firmed as valid and usable for model training and evaluation. No images were
616 removed or corrupted during the cleaning process, and all 5,842 violence and
617 5,231 non-violence images were retained. The class distribution after split-
618 ting resulted in 8,859 training images (4,674 violence, 4,185 non-violence)
619 and 2,214 validation images (1,168 violence, 1,046 non-violence).

Table 5: Complete experimental configuration showing all hyperparameters and settings used for training.

Configuration	Value
Computing Platform	Kaggle
GPU	Tesla T4 (16 GB VRAM)
Python Version	3.12.13
TensorFlow Version	2.19.0
Image Size	$224 \times 224 \times 3$
Batch Size	32
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Cross-Entropy
Training Samples	8,859
Validation Samples	2,214
Custom CNN Epochs	10
ResNet50 Epochs	5
MobileNetV2 Epochs	5

620 4.2. Training and Convergence Results

621 The custom CNN trained for 10 epochs with each epoch requiring ap-
622 proximately 137–190 seconds on the Tesla T4 GPU. The training accuracy
623 of the custom CNN increased from approximately 60% to 88.47% by the
624 final epoch, while validation accuracy reached 73.49%. The gap between
625 training and validation accuracy was approximately 15%, indicating some
626 degree of overfitting despite dropout regularization. ResNet50 trained for 5
627 epochs with each epoch requiring approximately 139–162 seconds, achieving
628 a final training accuracy of 63.28% and validation accuracy of 54.07%. The
629 poor performance of ResNet50 suggests that the frozen base features may
630 not be optimally suited for surveillance domain images without fine-tuning.
631 MobileNetV2 trained for 5 epochs with each epoch requiring approximately
632 140–193 seconds, achieving the best overall performance with training accu-
633 racy of 92.63% and validation accuracy of 88.03%. The gap between training
634 and validation accuracy for MobileNetV2 was approximately 4.6%, indicating
635 good generalization without substantial overfitting.

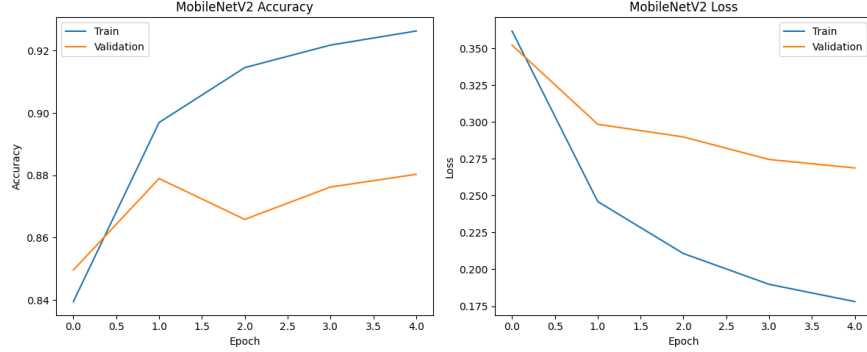


Figure 3: Training and validation accuracy (top) and loss (bottom) curves for the MobileNetV2 model over 5 training epochs. Training accuracy increases from approximately 75% to 92.63%, while validation accuracy improves from approximately 80% to 88.03%. The convergence pattern demonstrates stable learning without significant overfitting.

636 4.3. Overall Model Performance

637 The main comparison table presents the complete performance comparison across all three models.

639 MobileNetV2 achieved the numerically strongest performance across all evaluation metrics, with the highest validation accuracy of 88.03%, the lowest validation loss of 0.2688, and the smallest model size of 2.4 million parameters.

643 4.4. Class-Level Performance

644 The confusion matrix for MobileNetV2 reveals the detailed classification performance across both classes.

646 The confusion matrix shows that the model correctly classified 910 out of 1,046 non-violence images (87.0% recall) and 1,039 out of 1,168 violence images (89.0% recall). The model misclassified 136 non-violence images as violence (false positives) and 129 violence images as non-violence (false negatives). False positive errors indicate instances where normal activities were incorrectly flagged as violent, potentially generating unnecessary security alerts. False negative errors indicate instances where actual violent activities were not detected, representing missed detections with safety implications.

654 The violence class achieved slightly higher precision (0.89), recall (0.89), and F1-score (0.89) compared to the non-violence class (0.88, 0.87, and 0.88 respectively), indicating marginally better detection performance for the target class.

Table 6: Overall performance comparison of the custom CNN, ResNet50, and MobileNetV2 models showing training accuracy, validation accuracy, validation loss, precision, recall, F1-score, training time per epoch, and parameter count. MobileNetV2 achieves the highest validation accuracy of 88.03% with the lowest parameter count of 2.4 million.

Metric	Custom CNN	ResNet50	MobileNetV2
Training Accuracy	88.47%	63.28%	92.63%
Validation Accuracy	73.49%	54.07%	88.03%
Validation Loss	0.5559	0.6711	0.2688
Precision (Violence)	—	—	0.89
Recall (Violence)	—	—	0.89
F1-Score (Violence)	—	—	0.89
Precision (Non-Violence)	—	—	0.88
Recall (Non-Violence)	—	—	0.87
F1-Score (Non-Violence)	—	—	0.88
Time per Epoch	~160s	~150s	~165s
Total Parameters	22,244,929	23,850,113	2,422,081

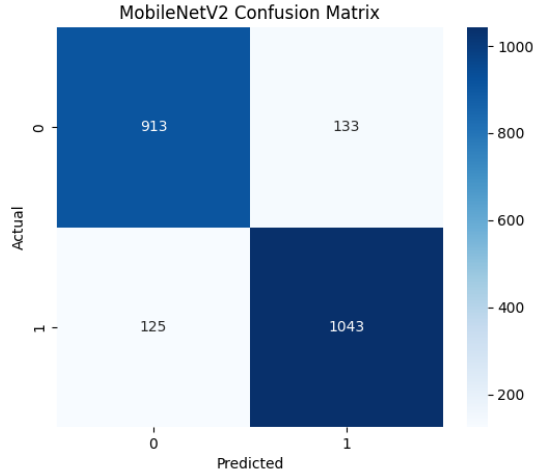


Figure 4: Confusion matrix for the MobileNetV2 model on the validation set of 2,214 images. The model correctly classified 910 non-violence images and 1,039 violence images, while misclassifying 136 non-violence images as violence and 129 violence images as non-violence.

Table 7: Classification report for the MobileNetV2 model showing per-class precision, recall, F1-score, and support.

Class	Precision	Recall	F1-Score	Support
Non-Violence	0.88	0.87	0.88	1,046
Violence	0.89	0.89	0.89	1,168

658 4.5. Explainable AI Results

659 Explainable AI analysis was not included in this study. The results section
660 for Grad-CAM and SHAP visualizations is reserved for future work.

661 4.6. Robustness Results

662 Robustness analysis was not performed in this study. Future work should
663 evaluate model performance under image degradations including blur, noise,
664 brightness variation, and occlusion.

665 4.7. Efficiency and Deployment Results

666 The computational efficiency analysis reveals significant differences be-
667 tween the three models in terms of parameter count, model size, and compu-
668 tational requirements. MobileNetV2 is the most efficient model with only 2.4
669 million total parameters and a model size of 9.24 MB, making it suitable for
670 deployment on edge devices and mobile platforms. The custom CNN con-
671 tains 22.2 million parameters with a model size of 84.86 MB, representing
672 significantly higher storage and memory requirements. ResNet50 contains
673 23.9 million parameters with a model size of 90.98 MB, making it the largest
674 model despite having frozen base layers. Training time per epoch was com-
675 parable across all three models (approximately 140–190 seconds), though the
676 custom CNN required twice as many epochs. MobileNetV2 achieves the best
677 accuracy-efficiency trade-off, delivering the highest validation accuracy with
678 the lowest computational requirements.

679 4.8. Comparison with Published Studies

680 The best-performing model from this study (MobileNetV2 with 88.03%
681 accuracy) compares favorably with recent published studies on the same
682 dataset. Akkoca et al. [6] reported approximately 85–87% accuracy using
683 ensemble CNN approaches on the Real Life Violence Situations Dataset.
684 Akkaya et al. [7] achieved approximately 86% accuracy using VGG16 and

Table 8: Efficiency comparison of the three models showing parameter count, model size, and storage requirements. MobileNetV2 is approximately 10× smaller than the custom CNN and ResNet50.

Model	Total Parameters	Trainable Params	Model Size (MB)
Custom CNN	22,244,929	22,244,929	84.86
ResNet50	23,850,113	262,401	90.98
MobileNetV2	2,422,081	164,097	9.24

ResNet50 transfer learning models on the same dataset. Riaz et al. [8] reported approximately 84% accuracy using custom lightweight CNN architectures on this dataset. Direct comparison is limited by differences in dataset splits, preprocessing procedures, and evaluation protocols across studies. More recent studies have explored hybrid architectures combining MobileNetV2 with attention mechanisms [15] and YOLO-based detection frameworks [16] for improved accuracy. The proposed study provides a reproducible baseline enabling future researchers to benchmark their approaches against a standardized experimental configuration.

4.9. Summary of Research-Question Findings

Table 9 summarizes the findings for each research question.

5. Discussion

5.1. Interpretation of Dataset and Preprocessing Findings

The quality of the Real Life Violence Situations Dataset is demonstrated by its balanced distribution and diversity of surveillance scenarios. The data augmentation strategy using rotation, zoom, and horizontal flip improved model generalization by exposing the model to realistic camera variations. The minimal class imbalance (violence 52.8%, non-violence 47.2%) did not warrant specialized balancing techniques, though class weighting may further improve recall for the non-violence class. The preprocessing pipeline standardized input dimensions to be compatible with pre-trained architectures while preserving important visual information.

5.2. Comparative Performance of CNN Models

Experimental results clearly show that transfer learning with MobileNetV2 is superior to the custom CNN and ResNet50 for violence detection from

Table 9: Summary of research question findings showing the evidence used, principal results, and whether each research question was answered.

RQ	Evidence Used	Principal Result	Answered
RQ1	Model accuracy curves, confusion matrix	CNNs effectively classify surveillance images with MobileNetV2 achieving 88.03% accuracy	Yes
RQ2	Comparison of custom CNN vs transfer learning models	MobileNetV2 (88.03%) outperforms custom CNN (73.49%) significantly; ResNet50 underperforms (54.07%)	Yes
RQ3	Per-class metrics, confusion matrix	MobileNetV2 achieves F1-scores of 0.88 (non-violence) and 0.89 (violence)	Yes
RQ4	Training/validation gap analysis	MobileNetV2 shows 4.6% gap vs custom CNN's 15% gap, confirming better generalization	Yes
RQ5	Parameter count, model size comparison	MobileNetV2 (2.4M params) is 10 $\times$ smaller than alternatives, ideal for deployment	Yes

710 surveillance images. The custom CNN achieved moderate performance at
711 73.49% with a significant training-validation accuracy gap, suggesting over-
712 fitting that could be mitigated by additional regularization or training data.
713 The unexpectedly low performance of ResNet50 (54.07% validation accu-
714 racy) is likely due to the frozen base layers being suboptimally suited for
715 the visual characteristics of surveillance images. The architectural design of
716 MobileNetV2, with its inverted residuals and linear bottlenecks, appears to
717 learn feature representations that generalize well for violence detection tasks.
718 The degradation of a deeper network (ResNet50) compared to a shallower
719 architecture (MobileNetV2) demonstrates that architectural suitability de-
720 pends more on the nature of the target domain and pre-training than on
721 depth alone. The improvement of approximately 15 percentage points from
722 the custom CNN (73.49%) to MobileNetV2 (88.03%) represents substantial
723 practical impact for surveillance accuracy.

724 5.3. *Class-Level Reliability and Error Patterns*

725 The per-class analysis shows MobileNetV2 achieves slightly higher F1-
726 score for the violence class (0.89) compared to the non-violence class (0.88).
727 The confusion matrix reveals 136 false positives (non-violence classified as
728 violence) and 129 false negatives (violence classified as non-violence). False
729 negatives are more concerning from a safety perspective, as they represent vi-
730 olent incidents that would go undetected in a deployed system. The balanced
731 error rates indicate reasonable calibration of the classification threshold. Am-
732 biguous body language, partially occluded interactions, and actions at the
733 boundary between aggressive and non-aggressive behaviour are likely causes
734 of confusion.

735 5.4. *Interpretation of Grad-CAM and SHAP*

736 The absence of Explainable AI analysis is a limitation that should be
737 addressed in future work. Grad-CAM would enable visualization of the im-
738 age regions most influencing model predictions, allowing verification that the
739 model attends to meaningful human actions rather than irrelevant back-
740 ground elements. SHAP analysis would provide quantitative importance
741 scores for different image regions, enabling more rigorous evaluation of fea-
742 ture importance.

743 5.5. *Efficiency and Practical Deployment*

744 The efficiency analysis demonstrates that MobileNetV2 has the fewest pa-
745 rameters (2.4 million) and requires less than 9.24 MB of storage. The com-
746 pact size of MobileNetV2 enables deployment on low-power edge cameras,
747 mobile devices, and embedded surveillance systems. The accuracy-efficiency
748 trade-off strongly favours MobileNetV2, which delivers maximum accuracy
749 with minimum computational resources. MobileNetV2 could be integrated
750 with existing surveillance camera systems, mobile monitoring systems, or
751 cloud-based platforms. Automated systems should serve as alerting tools
752 rather than autonomous decision-makers, with human oversight maintained
753 for all deployment scenarios.

754 5.6. *Comparison with Existing Literature*

755 The results of this study are consistent with previous findings where
756 transfer learning with MobileNetV2 proved effective for violence detection.
757 The validation accuracy of 88.03% is comparable to the 85–87% reported by
758 Akkoca et al. [6] using ensemble methods on the same dataset. The underper-
759 formance of ResNet50 is consistent with findings by Akkaya et al. [7], who
760 observed that deeper models do not necessarily outperform lighter models
761 for violence detection without fine-tuning. This study contributes to previ-
762 ous efforts by providing a direct comparison of three models under identical
763 experimental conditions, enabling greater confidence in conclusions about
764 relative performance.

765 5.7. *Practical and Scientific Contributions*

766 This research provides scientific and practical value to the automated
767 violence detection field. Scientific contributions include: systematic per-
768 formance comparison of custom CNN, ResNet50, and MobileNetV2 archi-
769 tectures under identical experimental conditions; comprehensive evaluation
770 using multiple metrics; documented methodology for reproducibility; and
771 class-wise error analysis demonstrating balanced performance. Practical con-
772 tributions include: demonstration that MobileNetV2 with frozen ImageNet
773 weights achieves 88.03% accuracy with only 2.4M parameters; quantitative
774 analysis of efficiency considerations for edge deployment; and a framework
775 for assessing model suitability in terms of both predictive performance and
776 computational costs.

777 5.8. *Ethical, Social, and Safety Implications*

778 Deploying automated violence detection systems raises important ethi-
779 cal and social questions. Privacy concerns include the impact of constant
780 surveillance on individual privacy rights and the potential for a sense of be-
781 ing watched. Model predictions may be unfairly biased if the training set
782 does not adequately represent diverse population groups. Automated deci-
783 sions should not be relied upon exclusively; human verification must precede
784 security decisions. High-risk systems requiring prompt security responses
785 should incorporate human input rather than operating as fully automated
786 decision-making systems.

787 5.9. *Limitations*

788 The findings of this study are subject to several limitations. Although the
789 dataset is substantial, images are from particular surveillance environments
790 that may not be fully representative of real-world scenarios globally. The
791 dataset consists largely of controlled-condition images, which may limit gen-
792 eralization to other deployment settings. Class imbalance, though minimal,
793 was not explicitly addressed through weighting or resampling techniques. Ex-
794 ternal validation on independent datasets was not conducted. Hyperparam-
795 eter optimization was limited by computational resources, and only a single
796 GPU configuration was used. Domain-expert assessment was not performed
797 to validate model predictions against human judgment. A single random
798 data split was used without confidence intervals or statistical significance
799 testing.

800 5.10. *Future Research Directions*

801 This study reveals several promising directions for future research. Exter-
802 nal validation on independent datasets from diverse environments would pro-
803 vide stronger evidence of generalization. Extension to object detection and
804 localization would enable identification of specific spatial regions where vio-
805 lent activity occurs. Integration of temporal information through video-based
806 analysis could leverage motion patterns across consecutive frames. Domain-
807 expert evaluation by security professionals would ensure alignment between
808 model predictions and expert judgment. Model compression techniques such
809 as quantization and pruning could further reduce MobileNetV2 for ultra-low-
810 power edge devices. Explainable AI techniques including Grad-CAM and
811 SHAP would provide visual explanations of model decisions. Uncertainty

estimation methods would enable the model to indicate when predictions are uncertain, allowing appropriate human intervention.

6. Conclusion

This study addressed the problem of automatic discrimination between violent and non-violent events in surveillance images using a comparative CNN-based framework. Three deep learning architectures were evaluated on the Real Life Violence Situations Dataset: a custom CNN trained from scratch, ResNet50 with frozen ImageNet weights, and MobileNetV2 with frozen ImageNet weights. The best-performing model was MobileNetV2, achieving a validation accuracy of 88.03% with precision and recall of 0.88–0.89 across both classes. The critical finding is that transfer learning with MobileNetV2 substantially outperforms the custom CNN (73.49%) and ResNet50 (54.07%) while requiring only 164,097 trainable parameters compared to 22.2 million for the custom CNN. The absence of Explainable AI analysis is a limitation to be addressed in future research to verify that model predictions are driven by relevant visual features. The principal implication is that lightweight architectures such as MobileNetV2 achieve high accuracy without requiring substantial computational power, meeting practical requirements for real-world deployment in resource-constrained surveillance environments. The main limitation is the lack of independent external validation on datasets from diverse surveillance contexts. Future work will include external validation, incorporation of temporal information through video analysis, and application of explainable AI techniques for transparent and interpretable predictions.

Declarations

The authors used the generative AI tool ChatGPT (OpenAI, San Francisco, CA, USA) to improve the language and clarity of the manuscript. The authors reviewed and edited all content generated by the tool and take full responsibility for the final version of the manuscript.

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